

Global Facilities – Design & Construction Standard – I&C – Facility Management and Control System (FMCS)

ECN Facility:	GLOBAL FACILITIES	ECN Area:	ENGINEERING
Owner(s)	ENGINEERING	Target Audience:	All Facilities
Document Type:	Standard	Document ID (pdf):	A3YRXSD74VDV-57553043-401
Current ECN:	101165330	Last Modified:	5/17/2024
Version:	3		

[ECN History](#)

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1.0 Purpose

This standard defines the minimum requirements for design, supply, and installation of Facility Management and Control System to greenfield project, including SCADA, PLC, system integration, cabling, cabling, factory & site testing, and commissioning.

2.0 Scope

2.1 Site(s) impacted

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

2.2 Vendor Scope

2.2.1 The following are in the scope of FMCS system integrator:

- System Architecture drawing (PLC, SCADA, Network Switches, etc).
- TCP/IP communication interface list.
- The layout drawings for cable routing, instrument and panel location, and zero reference pressure piping
- I/O list
- Control panel design drawings

- f. Cable schedule complete with all details such as Type, Size, Length, quantity of all cables.
 - g. Sequence of operation with FMCS for Micron approval.
 - h. Supply and install all necessary hardware, software, system network, cables, alarm devices and field instrumentation to build the FMCS.
 - i. Provide all the necessary services for designing, manufacturing, supply, delivery, installation, testing, commissioning, and handover.
 - j. Provision of Clean room / Hazardous area type devices (if applicable).
 - k. Liaison with other system vendors for necessary interfacing.
 - l. Program Software.
 - m. Safety Integrated Level (SIL) calculations if applicable.
 - n. Inspection and Test Plan (ITP).
 - o. Testing and Commissioning.
 - p. System Training.
 - q. Operation & Maintenance Spares.
 - r. As-Built Drawings.
 - s. Operation & Maintenance Manual.
 - t. 24/7 Monitoring of system at control room from system commissioning to handover of the system to Micron.
- 2.2.2 The FMCS shall be designed and installed to operate under the project site conditions.
- 2.2.3 The equipment and software proposed by the vendor shall be currently in manufacturer. No custom products shall be allowed. All products shall be supported by manufacturer for a minimum of ten (10) years from date of acceptance, including spare parts, board repairs and software revisions.
- 2.2.4 The scope shall include any special tools required for installation, field testing, commissioning, operation and maintenance of the FMCS.
- 2.2.5 Where required, commissioning spare parts are to be provided during commissioning of the system& equipment by the FMCS system integrator.
- 2.2.6 The FMCS system integrator shall include as optional, recommended 2 years operational spares.
- 2.2.7 FMCS system integrator shall ensure that all equipment and materials are designed and packed for transit to site without damage.

- 2.2.8 The FMCS system integrator shall ensure a proper organization structure for the project development and management. The system integrator shall submit list of persons and resume of everyone who will perform project manager, engineer, and construction management functions in the project.
- 2.2.9 The FMCS system integrator shall deliver products or services as per requirement specification and shall have an appropriate quality management system.
- 2.2.10 FMCS SCADA system integrator shall be Wonderware certified system platform system integrator with minimum past 5 years Wonderware system platform project experience.

3.0 Definitions and Acronyms

- **A/D** – Analog to Digital
- **AC** – Alternating Current
- **AI** – Analog Input
- **AO** – Analog Output
- **D/A** – Digital to Analog
- **DC** – Direct Current
- **DI** – Discrete Input
- **DO** – Discrete Output
- **EHS** - Environmental, Health, and Safety
- **FAT** – Factory Acceptance Test
- **FCC** – Fire Command Center
- **FMCS** – Facility Management and Control System
- **I&C** – Instrumentation and Control
- **ITP** – Inspection and Testing Plan
- **HAZOP** - Hazard and Operability
- **O&M** – Operation and Maintenance
- **OPC** – Object linking and embedding for Process Control
- **PLC** – Programmable Logic Controller
- **PC**– Personal Computer
- **PID** – Proportional Integral and Derivative control algorithm
- **RIO** – Remote I/O
- **SAT** – Site Acceptance Test
- **SCADA** – Supervisory Control and Data Acquisition
- **SI** – System Integrator
- **T&C** – Testing and Commissioning

4.0 Roles and Responsibilities

Roles	Responsibilities
Global Facilities Engineering Team (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
Global Facilities Regional Team (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from Design Standard • Share new lessons learned to both Global Engineering and Regional Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of Design Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Applicable Codes and Standards

The following industry, association, codes and standards shall be followed as applicable to the design, fabrication, assembly, and testing of all equipment furnished under this specification. Deviations from this must be agreed formally by Micron.

- a. American National Standards Institute (ANSI)
- b. Institute of Electrical and Electronics Engineers (IEEE)
- c. IEEE 1985, Guidelines for the Documentation of Software in Industrial Computer Systems.
- d. National Electrical Manufacturer Association (NEMA)
- e. National Electrical Code (NEC)
- f. Instrument Society of America (ISA)
- g. Good Automated Manufacturing Practices (GAMP)
- h. National Fire Protection Association (NFPA)

- i. International Electrotechnical Commission (IEC): IEC 61131-3, IEC 60529, IEC 60255, IEC, 60664, IEC 60668.
- j. ISO 9001 – Quality Management Systems
- k. Underwriters Laboratories (UL): UL 508 and UL 508A.
- l. [Global EHS – EMO Button Guideline.](#)

Electrical and instrument equipment and materials for installation should comply with hazardous level requirement. Refer to NFPA-496.

6.0 System Description

- 6.1 The FMCS includes a layered architecture of devices with functionality required for continuous operation, monitoring, and administration of facility mechanical and process systems.
- 6.2 The FMCS system consist of PLCs, I/O, HMI/SCADA equipment required to control and monitor facilities equipment. Components typically connected to the FMCS are instrumentation, switches, valves, VFDs, and I/O.
- 6.3 The Supervisory Control and Data Acquisition (SCADA) level is comprised of data collection, data processing, and communication servers connected by an Ethernet network.
- 6.4 The primary control level is comprised of distributed PLCs and local or Remote I/O (RIO) interface equipment.
- 6.5 Separate vendor provided SCADA systems will communicate with the FMCS using Object Linking and Embedding for Process Control (OPC).
- 6.6 In general, based on complexity of the facility system being monitored, the SCADA / HMI shall incorporate graphical representation of systems to display status and alarms. Graphics shall be representative of, but are not limited to, Piping and Instrument Diagrams (P&ID's), Process Diagrams (PFD's), I/O list, wiring schematics and/or instrument loop diagrams.
- 6.7 The FMCS will run on multi-user, multitasking operating system using remote terminals and/or personal computers as operator interfaces.

7.0 Design Requirement

- 7.1 The FMCS system shall be designed and developed with, but not limited to, P&ID, Sequence of Operation, I/O list and Control System Architecture.
- 7.2 The FMCS system shall utilize PLC, SCADA, integration, and networking.
- 7.3 Redundant PLC processors and remote I/O should be considered when they control life safety functions or if system failure puts personnel at an unnecessary risk or critical processes.
- 7.4 Latching relays should be used on equipment, where appropriate, to minimize equipment downtime in the event of a PLC failure.
- 7.5 All PLC and remote I/O panels shall have redundant power supplies.
- 7.6 Each incoming power to control panel is monitored with alarm enable.
- 7.7 All PLC and remote I/O panels shall be built to UL508A or equivalent specifications.
- 7.8 A common PLC platform and HMI platform should be utilized to reduce cost of ownership.
- 7.9 PLCs shall allow for the configuration and addition of I/O, as well as program downloads in run mode.
- 7.10 All PLCs and remote I/O shall be provided with short circuit protection.
- 7.11 All PLC should incorporate IEC-61131-3 programming standard.
- 7.12 All PLCs shall provide the ability to force all I/O.
- 7.13 All PLCs shall have Ethernet capable communications.
- 7.14 All PLC and remote I/O shall provide the ability to remove and replace modules while the system is still powered (hot-swappable).
- 7.15 All PLC and remote I/O cabinets shall have cooling fans.
- 7.16 All PLC and remote I/O power supplies shall be monitored for loss of power.
- 7.17 All PLC and remote I/O shall be monitored for loss of communications between components.
- 7.18 All PLC and remote I/O shall have diagnostic capabilities that are capable of alarming to control room.
- 7.19 All PLC shall have the capability for on-line program modifications without shutting down the process under control.

- 7.20 All I/O points shall have current limiting circuitry to prevent damage from shorts in the field wiring.
- 7.21 Whenever possible, the instruments shall be loop powered.
- 7.22 Additional I/O should be supplied to include 20% spares and 20% additional rack and terminal block capacity.
- 7.23 All FMCS alarms shall report to a constantly attended centralized control room.
- 7.24 Data shall be stored to an SQL server at 1 to 5 seconds intervals.
- 7.25 The SCADA/HMI should incorporate web-based solutions that allow the system to retrieve SOO, drawings, and other important information from various system web pages.
- 7.26 Alarm should be generated at the PLC layer or level, rather than the SCADA level, to provide the ability to automatically perform a control response or function if required by the alarm. This will also ensure that when the SCADA system is down, and automatic control response of function can be conducted without input from the SCADA system.
- 7.27 Annual maintenance shall be performed on all PLC and remote I/O cabinets.
- 7.28 Maintenance shall include, but is not limited to, checking power supplies, checking/replacing batteries, checking status lights, and checking air supplies.
- 7.29 When performing maintenance within a panel, personnel shall use a grounding strap to prevent static charges that may damage equipment.
- 7.30 All materials shall be listed by UL or by equivalent listing agency.
- 7.31 A spare components inventory plan shall be available to have components either on-site or quickly available through manufacturer / manufacturer's representative. The volume and required components of the inventory shall be coordinated between the manufacturer, operations personnel and site controls engineering.
- 7.32 The site shall have personnel trained to replace PLC modules or have an agreement in place with the manufacturer / manufacturer's representative to provide personnel to replace components and/or provide technical support upon their failure.

8.0 Communication and Network Infrastructure

- 8.1 The facilities control network should be isolated from I.S network to eliminate exposure to I.S events and provide security to the control network.
- 8.2 The communication network is accomplished in the following manner: PLC to SCADA via Ethernet through VLAN; PLC to PLC via Ethernet through switches; PLC to RIO via industrial communication protocol. The network is capable for peer-to-peer (PLC-PLC) and programming computer-to-PLC communications via the network.
- 8.3 Configure the system to achieve a minimum overall system data throughput performance of 256 16-bit registers read from each PLC in a maximum time of 2 seconds when initiated by an operator. Achieve a minimum data throughput of writing 128 16-bit registers to any PLC from the SCADA system in a maximum time of 1 seconds.
- 8.4 Provide 30% growth capability in each PLC network without compromising data throughput capacity specified at full build out.
- 8.5 Provide a network architecture minimizing interface boxes, bridges, and similar hardware.

9.0 I/O List Requirement

The I/O list shall have, but not limited to, the following information to assign each I/O:

- a. SCADA point tag
- b. Building/Area
- c. Equipment Type or Instrument Type
- d. Equipment Tag or Instrument Tag
- e. Equipment or Instrument Description
- f. Power supply requirement
- g. P&ID / drawing / document reference.
- h. PLC/RIO Panel Tag
- i. PLC/RIO Panel location
- j. I/O Type (Digital Input, Digital Output, Analog Input, Analog Output, Pulse Input, High Level (OPC, Modbus, Profibus, etc), Others – to be specified)

10.0 Components

10.1 PLC

- 10.1.1 The FMCS system shall utilize a Programmable Logic Controller (PLC) to provide monitoring and supervisory control functions.
- 10.1.2 Modular with capability for easy expansion and future hardware and software upgrades, distributed structure, extensive communication capabilities, and user-friendly programming software.
- 10.1.3 Support of multiple programming languages. IEC 61131-3 compliant.
- 10.1.4 Fault tolerant including redundant power supplies, CPUs, and network capabilities.
- 10.1.5 Conformance to current revisions of standards, but not limited to: ISO 9001, IEC 60664 and IEC 61000 (Isolation), IEC 60668 and IEC 60529 (Environmental), and IEC 60255.
- 10.1.6 On-line hardware, software, and communications diagnostics tools. The diagnostic messages of CPU, power module, communication, and I/O modules shall be sent to SCADA for monitoring and alarming. The status of each module and system communication status shall be monitored and displayed with its location in the SCADA.
- 10.1.7 Connectivity to SCADA field devices, Human Machine Interfaces (HMI).
- 10.1.8 Provide PLC capable of initiating data messages to allow data reporting by exception to be performed by the supervisory software.
- 10.1.9 Use copper or optical fiber media or both for network communications to optimize PLC network speed.
- 10.1.10 Design network with segments and limited number of PLCs per segment to optimize communications speed.
- 10.1.11 Straightforward and efficient installation, calibration, and maintenance.
- 10.1.12 Provide local and factory vendor technical support.
- 10.1.13 Conformal coating for environmental protection.
- 10.1.14 Operate within a temperature range of 32 to 140 degrees F (0 to 60 degrees C) and 5 to 95 percent of relative humidity (non-condensing) at 140 degrees F (60 degrees C).
- 10.1.15 Minimum MTBF requirements: 250,000 hours for processors; 500,000 hours for communication modules; 1,000,000 hours for I/O modules.

- 10.1.16 Retain PLC memory configuration for a minimum of 6 months without applied power.
- 10.1.17 Each controller shall be sized in accordance with the following requirements:
 - a. Additional I/O should be included to provide 20% extra spare points of each I/O type (AI, AO, DI, DO) for future expansion.
 - b. 20% spare space shall be available on control rack to allow the addition of future modules.
- 10.1.18 Provide blank cover plates for spare PLC slots.
- 10.1.19 Supply PLC racks capable of containing one or more communication modules to provide communication interfaces to third-party devices.
- 10.1.20 Provide PLC capable of establishing communications and exchange data, via the adequate modules installed on PLC backplane, with third-party equipment. The communication protocols include, but not limited to, Modbus, TCP/IP Ethernet interface, RTU protocol, ASCII, Profibus, and BacNet.
- 10.1.21 Provide PLC with at least one port to support local programming capability.
- 10.1.22 Provide PLC communication cables satisfying requirements of PLC manufacturer and approved by Micron. Install cable in accordance with installation requirement of the PLC manufacturer.
- 10.1.23 10.1.23 Time in PLC shall synchronize with FMCS network or GPS time.

10.2 I/O Interface

- 10.2.1 In every module I/O including spare will be fully wired.
- 10.2.2 I/O segregation: Signal lines from redundant sensors or equipment shall connect to different I/O modules in different rack.
- 10.2.3 Provide the ability to monitor and override I/O.
- 10.2.4 Provide I/O rack size configured for the base bid, suitable for panel mounting.
- 10.2.5 Provide the ability to pre-select the failure status of each output point in the event of CPU failure.
- 10.2.6 Provide PLC modules supporting factory-provided wiring interface devices which allow the removal and reinstallation of I/O modules without removing wires.
- 10.2.7 Provide ability to assign PLC I/O addresses regardless of physical location.
- 10.2.8 Discrete I/O Module requirement:
 - a. Fuse protect output channels. Include fuse with the PLC hardware and provide a blown fuse indication.
 - b. Source outputs at 24Vdc continuous service. Provide discrete outputs with relay outputs where required to interface to motor controls and other circuits requiring isolation.
 - c. Provide individual status lights for each input and output. Provide status indicating light (okay or fault) for each module.
- 10.2.9 Analog I/O Module requirement:
 - a. Provide analog input modules capable of accepting 4 to 20mA dc with an input impedance of 250 ohms.
 - b. Provide analog output modules capable of driving a 4 to 20mA dc signal into a zero to 600 ohms load.
 - c. Provide A/D and D/A conversion with a resolution of 12 bits minimum. Accuracy, linearity and repeatability shall be 0.1% or better. Ambient temperature effect shall be less than 0,01% per °F.
 - d. Fuse protect output channels. Include fuse with the PLC.
 - e. Analog I/O points shall be individually isolated. Analog inputs and outputs shall be individually fused.
 - f. The following types of analog I/O shall be supported by the system without the use of external signal conditioning.
 - i 4-20 mA input, loop and externally powered;
 - ii Millivolt;
 - iii RTD;

- iv Thermocouple;
- v Frequency;
- vi 4-20 mA output.

10.2.10 The following types of digital I/O shall be supported by the system. The vendor shall specify when interposing relays are required and whether or not there are provisions for them on the Vendor's standard termination board:

- a. Dry Contact input, system powered;
- b. Relay output, 117 VAC @ 2amps, minimum;
- c. Voltage input, 24 VDC;
- d. Voltage input, 117 VAC;
- e. Voltage output, 24VDC, 115VAC.

10.2.11 Digital inputs shall be individually optically isolated. Digital inputs and outputs shall be individually fused.

10.2.12 When sizing, selecting, and coordinating fusing for the FMCS system, the following factors shall be considered:

- a. Normal operating current – The current rating of a fuse should typically be de-rated 25% for operation at 25 °C to avoid nuisance fuse failure.
- b. Application – The voltage rating of the fuse shall be equal to or greater than the available circuit voltage.
- c. Ambient Temperature – Derating the fuse shall be considered if the ambient temperature is greater than 25 °C.
- d. Overload current and length of time in which the fuse must open.
- e. Maximum available fault current – the interrupting rating of a fuse must meet or exceed the maximum fault current for the circuit.

10.2.13 All fusing shall be coordinated to ensure that an overcurrent condition at the instrumentation level will not affect the power distribution to other instruments, PLC modules, or power supplies.

10.3 SCADA

- 10.3.1 SCADA application servers shall consist of redundant pairs of servers that have the ability to failover to one another. Recommended uptime goal is 100%.
- 10.3.2 SCADA I/O servers shall be configured such that if one fails, there is sufficient capacity in another server to assume the load.
- 10.3.3 The SCADA shall continuously monitor the facility equipment operation and activate alarms.
- 10.3.4 To define latest server hardware specification based on highend server capability and IT requirement.
- 10.3.5 The SCADA system shall have redundant historical server with capacity to retain historical data for minimum 2 years of up to 5 seconds data capacity.
- 10.3.6 The SCADA system shall be engineering workstation to program and document the SCADA application software.
- 10.3.7 The SCADA Software requirement:
 - a. The SCADA system shall be based on Wonderware Galaxy system platform.
 - b. Provide device alarming including communication.
 - c. Use object or template-oriented programming when possible.
 - d. Historize alarming for minimum 2 years to allow of troubleshooting and trending.
 - e. Provide communication monitoring/alarming.
 - f. Keep servers up to date with security patches and operating systems patches and updates.
 - g. When the system is down, sites should have procedures in place to provide manual monitoring and data logging until system is restored.
 - h. The alarm limits in the FMCS are variable. These alarm limit parameters may only be changed via audit trail by an authorized person protected by adequate access control and system privileges. All changes are logged.
 - ix. Alarm indication and handling shall be provided. Alarms will be logged in an audit trail. Alarm ranges will be configurable. Sorting and filtering of alarms will be possible.
 - i. The FMCS shall provide authorized system access for predefined user groups like System Administrator, Supervisor, and User. Individual system functions will be assigned or limited to different user groups.
 - xi. Creation, modification, and deletion of monitoring parameter settings, alarm limits, shall be restricted to authorized personnel and shall be controlled.

- j. Modification and deletion of monitored raw data must be prevented. Monitoring raw data must be protected. xiii. Historical Data shall be retained for minimum 2 years. xiv. The SCADA/HMI should incorporate web-based solutions that allow the system to retrieve SOO, drawings, and other important information from various system web pages.

10.3.8 To ensure standardization of graphic screens, all sub-system SCADA are required to use the same standards for graphics, alarming, trending, and other applications' programming. The standards specified in this guideline shall be used as the minimum requirement.

10.3.9 Mechanical and process system graphics shall follow process flow diagram, P&ID, and layout drawings.

10.3.10 The SCADA graphic faceplate samples are available in Appendix B for guideline.

10.4 PLC Programming Tool

- a. Provide software and hardware interfaces (engineering workstations) required to program and document the PLC application software.
 - i Capable of programming PLCs from a central location, FMCS control room.
 - ii Run on a multitasking personal computer.
 - iii Capable of performing on- and off-line programming, support upload and download operations, and monitoring and editing of PLC data and programs online.
 - iv Ability to modify and edit the PLC program and download changes with the PLC I/O holding last state. ii. Supply and install programming software on a laptop computer, capable of interfacing with the PLC locally.

10.5 Third Party Interface

- a. Use proven off-the-shelf 3rd party communication software, i.e. TopServer, Kepware, OmniServer.
- b. Use industry-proven protocols in lieu of vendor-based proprietary protocols.
- c. Integrate error checking communication programming to accommodate vendor protocols where available, i.e. check summing and error code response.

10.6 Control Panel

- 10.6.1 Fabricate control enclosures 48 inches by 36 inches or smaller suitable for wall or a strut rack mounting.
- 10.6.2 Fabricate control enclosures larger than 48 inches by 36 inches as freestanding panels fitted with removable lifting lugs and legs minimum 12 inches high. Provide steel nuts and bolts with gaskets to fill holes after removal of lifting lugs.
- 10.6.3 Control Panels and Enclosures:
 - a. Standard size prefabricated enclosures manufactured by Hoffman or Rittal or approved equal.
 - b. Enclosures will be NEMA 4X for wet, damp and corrosive locations; NEMA 12 for all other locations.
 - c. Oil-tight latching mechanisms with keyed-alike locking handle. Tape keys to inside of doors.
 - d. Gasketed overlapping front access doors, with heavy gauge hinges and three-point latch rods with rollers.
 - e. Internal LED panel light with safety cover, end cap, and convenient on/off switch. Provide one fixture for each 48 inches of panel width.
 - f. 9 inches by 12 inches by 1-inch print pocket to store arrangement, schematic, and loop drawings.
 - g. Minimum 14-gauge steel.
 - h. Electrical power sockets that are isolated from the control wiring.
 - i. Cooling package (e.g. fan exhaust, enclosure air conditioner coolers) sized for published heat rejection information for powered components. Incorporate integral filters and sealing gaskets into fans and exhaust vents without compromising the panel environmental rating.
 - j. Install enclosure heaters on enclosures that will not be located within climate-controlled spaces to prevent condensation and resulting corrosion.
- 10.6.4 All field wiring shall be bottom entry.
- 10.6.5 Provide wiring between terminal blocks and panel equipment.
- 10.6.6 Fasten components (e.g. terminal blocks) to interior surfaces of panels. Mount operator interface components on front access door/panel.
- 10.6.7 Install copper grounding bus bar and studs, and provide 10% (minimum of 3) spare taps and screws for future and field grounding use.

- 10.6.8 Provide ground cable connection over hinge between subpanel and door.
- 10.6.9 Provide separate wireways for AC and DC signals. Separate AC wiring from DC wiring. Maintain a 4-inch minimum spacing between parallel runs of power and control wiring and electronic instrument wiring.
- 10.6.10 Size wireways so that total cross-sectional area of installed wire will not exceed 20% of cross-sectional area of wireway. Locate wireways to eliminate interference with access to panel-mounted instruments or devices. Do not encroach on space allotted for future instruments and do not install terminal strips and splices in wireways.
- 10.6.11 Group wires to panel-mounted instruments and switches through wireways or cable ties.
- 10.6.12 Provide finger-safe terminals and components per IP-20.
- 10.6.13 Cut, punch, or drill panel cutouts and smoothly finished with rounded edges.
- 10.6.14 Provide one-piece back panels.
- 10.6.15 Mount components to the back panel using machine screws or bolts with tapped holes according to manufacturer's recommendations. Provide lock washers with bolts and screws.
- 10.6.16 Mount components such that:
 - a. Adjustments can be made without the removal of instruments or interruption of service to adjacent components.
 - b. Their installation will not be affected by, or interfere with, the removal of panel instruments.
 - c. Components can be removed from the front without removal of the back panel.
- 10.6.17 Support door-mounted equipment such that no appreciable or noticeable sag occurs and no excessive wear on the hinges or wiring occurs.
- 10.6.18 Provide lockout/tagout capable main power switch with circuit breaker or fused disconnected and terminal block for termination to each AC power (120 or 240V) feeder circuits. Provide a power terminal block for termination of the wires.
- 10.6.19 Provide power distribution component layout and wiring in accordance with the Drawings. Provide individually protected Vac and 24Vdc circuits for instruments as required plus 20% for future.
- 10.6.20 Provide a circuit breaker panel with circuit breakers for each specific power source to accommodate panel lighting, power supplies, instruments, I/O modules, and processors.

- 10.6.21 Provide sufficient space to allow AC power cables to run directly from the power block to the circuit breakers.
- 10.6.22 Terminate internal power wiring so that an interruption at any one point will not disrupt power to remaining components.
- 10.6.23 Wiring for Front of Panel Devices:
 - a. Provide hinge wiring wrapped with plastic cable stays. Secure hinged wiring at each end so that any bending or twisting will be around the longitudinal axis of the wire. Protect the band area with a sleeve.
 - b. Attached wiring to interior side of panel door to house wires for panel front-mounted devices.
- 10.6.24 Terminate interconnecting wires, including spares, between panel-mounted equipment and external equipment at numbered terminal blocks. Multi-conductor cable is not allowed except for I/O module wiring harnesses.
- 10.6.25 Arrange wiring neatly, cut to proper length, and remove surplus wire. Provide abrasion protection for wire bundles, which pass through holes or across edges of sheet metal. Spliced or taped wiring, wire nuts, and butt splices are not acceptable.
- 10.6.26 Wire Labels:
 - a. Preprinted, slide-on type or heat shrink cover Brady type, which may be wrapped around wire.
 - b. Affix wire labels at both ends of wire.
 - c. Cloth or adhesive tape is not acceptable.
- 10.6.27 Install wiring in such a way to provide protection from inadvertent grounding and cross connection.
- 10.6.28 Service Legends and Nameplates: Engraved, rigid, laminated plastic type with adhesive back; White with black letters unless otherwise noted.
- 10.6.29 Firmly affix interior front of panel nameplates to the panel without compromise enclosure rating.
- 10.6.30 Firmly affix interior nameplates next to the device with industrial grade adhesive.
- 10.6.31 Comply with grounding requirement as per NFPA 70 or equivalent.

10.7 Cables

- 10.7.1 All cables shall be low smoke zero halogen (LSZH) flame retardant type and shall be of adequate cross sectional area.
- 10.7.2 Lightning surge protectors shall be provided for all signals sent by copper cables external to the building, with the following requirements:
 - a. Provides protection from voltage spikes, sags, and surges.
 - b. Provides minimum 40 dB reduction of electrical noise for frequencies from 50kHz to 50 MHz.
 - c. Complies with UL 1449 for surge withstand capability.
- 10.7.3 Where cables and conductors serve as communication links between controllers, provide surge-protection circuits at each end which meet IEEE Standard C62.37 surge withstand capability tests.
- 10.7.4 Wire and Cable products.
 - a. Instrumentation cable 300V rated for 480V MCC/VFD applications, single pair, overall shield, type PLTC:
 - Individual Conductors: 18 AWG, stranded, tinned copper, polyethylene or PVC insulated, rated 105 degrees C, color coding per client standard.
 - Assembly: Twisted single pair, continuous 100 percent coverage aluminum-polyester shield, 18 AWG, stranded, tinned copper drain wire, overall PVC jacket. Maximum outer diameter 0.28 inches.
 - b. Instrumentation cable 300V rated for 480V MCC/VFD applications, multiple pairs, overall shield, type PLTC:
 - Individual Conductors: 18 AWG, stranded, tinned copper, polyethylene or PVC insulated, rated 105 degrees C, black and white numerically printed and coded pairs.
 - Assembly: Individual twisted pairs having 100 percent coverage aluminum-polyester shield and 20 AWG, stranded, tinned copper drain wire. Provide conductor bundle with 100 percent coverage overall aluminumpolyester shield and 18 AWG drain wire. Isolate group shields from each other. Wrap bundle with cable tape and cover with PVC jacket with overall PVC jacket.
 - c. Ethernet Communications Cable: Industrial Ethernet
ENET IP 10GBPS cabling: Category 6A UTP.
Terminations: Shielded jack modules RJ45 IP20.

- d. Profibus DP Communications: Belden 3079A or Siemens equivalent, TSP 22 AWG Solid Bare Copper Conductors, Beldfoil (100% Coverage) + TC Braid Shield (65% Coverage), Foamed High-density Polyethylene Insulation, PVC Jacket, 300V 75°C
- e. Fiber Optics Cable : 62.5/125 or 100/140 Multimode with ST connectors.

10.7.5 Cable and Wire Installation:

- a. Install wires only in approved and complete raceways.
- b. Prevent mechanical damage to conductors during installation.
- c. Install and terminate all wire in accordance with manufacturer's recommendations.
- d. Neatly arrange and lace conductors in all panelboards, pull boxes, and terminal cabinets by means of wire ties. Do not lace wires in gutter or panel channel. Install all wire ties with a flush cutting wire toe installation tool, use a tool with an adjustable tension setting. Do not leave sharp edges on wire ties.
- e. Terminate solid conductors at equipment terminal screws with the conductor tightly wound around the screw so that it does not protrude beyond the screw head:
 - Wrap the conductor clockwise so that the wire loop is closed as the loop is tightened.
 - Do not use crimp lugs on solid wire.
- f. Thoroughly clean wires before installing lugs and connectors.
- g. Terminate stranded conductors on equipment box lugs such that all conductor strands are confined within the lug. Use ring type lugs if box lugs are not available on the equipment.
- h. Spare conductors are to be coiled and taped and mark as "SPARE".
- i. Install instrumentation class cables in separate raceways from power cables.
- j. Do not make intermediate terminations.
- k. All new wiring shall be tested for integrity and functionality after it is installed.

10.8 Terminal Blocks

- 10.8.1 Provide minimum of 10% spare terminals; identify spare terminals.
- 10.8.2 Space terminal strips so that future expansion, rework / maintenance, and field connections are accomplished without special tools.
- 10.8.3 Sectional Plastic Blocks:
 - a. DIN rail mounting with side entry.
 - b. Rated for 300 or 600 volts suitable the application.
 - c. Double sided.
 - d. Capable of accepting wire sizes up to AWG 12.
 - e. Permanent, legible identification.
 - f. Include end clamps and cover plates.
 - g. Provisions for ganged terminal jumpers and test disconnect jumpers.
- 10.8.4 Elevate terminal strips so the top of the smallest terminal on the strip is even with the top of the adjacent wireway.
- 10.8.5 Install no more than two wires and a prefabricated jumper in any terminal block.
- 10.8.6 Provide fused terminal blocks with fuses for any loop that is powered from the panel (e.g., analog I/O, non-isolated Digital outputs). Provide knife disconnect terminals for digital inputs and isolated digital outputs.
- 10.8.7 Provide grounded shield terminals for analog I/O field connections.
- 10.8.8 Acceptable manufacturers:
 - a. Weidmuller
 - b. Wago
 - c. Or approved equal.

10.9 Cable Tray / Trunking

- 10.9.1 Signal wires and sample tubing should not be installed in the same raceway.
- 10.9.2 Control cables should be routed in ELV cable tray / trunking.
- 10.9.3 Conduit type to be used is depending on area:
 - a. Rigid metal conduit and fittings (for outdoor/indoor, concealed/exposed, embedded in Concrete, Underground, EMI sensitive area, :
 - Rigid galvanized steel.
 - Rigid aluminum conduit.

- Rigid steel conduit with external 40-mil PVC coating and internal phenolic coating over a galvanized surface.
 - Fitting and conduit bodies: threaded type, steel/aluminum.
 - Hazardous area sealing fittings: UL-listed threaded type conduit seal fittings and sealing compound suitable for hazardous location installations in accordance with NEC 501-5.
- b. Intermediate Metal conduit (IMC): galvanized steel and use fittings and conduit bodies specified above for rigid steel conduit. Used for outdoor/indoor, concealed, embedded in concrete, underground,
- c. Electrical Metallic Tubing: Galvanized tubing. Used for outdoor/indoor, concealed.
- d. Liquid-tight flexible conduit: Flexible metal conduit with PVC jacket; fittings and conduit bodies: metal with PVC jacket. Used for final connection where flexible connection is required to minimize vibration or to facilitate equipment/instrument removal or adjustment. Maximum length is 1 meter.
- e. Rigid nonmetallic conduit: Schedule 40 PVC; fittings and conduit bodies: PVC. Used for outdoor/indoor, concealed, embedded in concrete, embedded in concrete, underground, corrosive nonhazardous area. vi. Liquid-tight nonmetallic flexible conduit: plastic; fittings and conduit bodies: PVC. Used for final connection where flexible connection is required to minimize vibration or to facilitate equipment/instrument removal or adjustment. Maximum length is 1 meter.
- 10.9.4 Furnish and install new conduit and/or wire-way as shown diagrammatically on the installation plan drawings. Provide all supports, brackets and mounting hardware as required.
- 10.9.5 Control cables tray shall be separated from power cable tray.
- 10.9.6 Accessories and Fittings: Manufacturer's standard clamps, hangers, brackets, splice plates, reducer plates, blind ends, barrier strips, connectors, and grounding straps.
- 10.9.7 Use bonding jumpers across expansion connectors for cable tray ground continuity. Provide bonding continuity between metallic cable tray sections and fittings in accordance with manufacturer's instructions. Make connections to aluminum tray and fittings using an antioxidant compound. Bond cable tray systems to the power system's ground every 50 feet or based on local code requirement. Do not use bare copper grounding conductors in aluminum cable trays.
- 10.9.8 Furnish and install approved Fire-stopping at all penetrations through rated fire walls/ceilings.

10.10 Power Requirement

- 10.10.1 All PLC and remote I/O panels that are controlling multiple facilities systems shall be powered with two (2) circuits with primary circuit shall be Emergency Backed UPS (essential) and the secondary circuit shall be Normal Power for the purpose of power failure and ability to perform maintenance; or dual emergency-backed UPS (essential) from different feed (different transformers).
- 10.10.2 PLCs and remote I/Os that control equipment that is skidbased or can stand alone, such as chemical systems, can be powered with one (1) circuit, however, the design shall be fail-safe to avoid risk to personnel and production.
- 10.10.3 All HMI/SCADA servers and control nodes shall be powered with emergency backed UPS (essential).
- 10.10.4 For redundant sensor at critical PID loop, use different power source.
- 10.10.5 All redundant DC power shall connect in parallel.

10.11 IT Requirement

- 10.11.1 The SCADA server and workstation will be coordinated with IT for integration.
- 10.11.2 PLC and SCADA shall be on Facility dedicated VLAN.

11.0 Commissioning

11.1 General Requirement

- a. Systematic process of ensuring systems perform interactively per Project design intent and Owner's performance requirements.
- b. Inspection and Test Plan (ITP) or T&C Responsibility Matrix should be submitted together with Quality management system for Micron approval during design engineering stage. The ITP shall include, but not limited to, FAT list, Instrument, Installation Inspection, Cable test, Point-to-Point test and SAT Functional test list and schedule.
- c. The commissioning can apply two tags: Green tag and Blue tag. The Green Tag and Blue Tag commissioning. Green Tag is when a green identification card is attached to each piece of equipment to signify the equipment has successfully conformed with design verification, and equipment or system and control readiness have been validated for start-up. Blue Tag is when a blue identification card is attached to a piece of equipment, system, sub-system to signify the equipment has successfully conformed with functional testing and is capable of being operational. The Green Tag and Blue Tag checklist is provided in

Appendix D for guideline.

- d. The organizing, coordination of personnel, and scheduling of all tests to be the responsibility of the System Integrator. Notification of testing to be submitted to each testing participants a minimum of 2 weeks prior to the requested testing date.
- e. Test programs and procedures to be created by the System Integrator. Test procedures to have signoff spaces for Micron and the System Integrator.
- f. The test report shall include the following requirement, but not limited to,:
 - Project identification
 - Personnel name, signature and date. The date shall be as when the report is signed.
 - List of equipment used in the test with a photocopy of calibration certificate.
 - Graphic or Trend Chart or historical report to support the test report.
- g. Pretest the system for satisfactory operation prior to Micron system acceptance testing. Prior to Micron representative witnessing any testing, a system test report demonstrating satisfactory pretest results to be submitted to Micron 2 days prior to the testing test date.

11.2 FAT (Factory Acceptance Test)

11.2.1 Factory Acceptance Test (FAT) requirements:

- a. FAT list shall be included in ITP or T&C Responsibility matrix.
- b. The PLC/RIO panel and equipment shall be inspected during manufacturing and before shipment for site installation. Prerequisite for shipping is the Micron approved FAT.
- c. The Vendor prepares the FAT procedures for approval of Micron.
- d. The FAT will consist of, but not limited to, the following checks:
 - Check conformance to Design Standard of Micron and Micron-approved technical submissions of the System Integrator.
 - Check for completeness.
 - Check for materials.
 - Verify correctness of nameplate data.
 - Verify all SIL certificate, IS certificate & explosion certificate etc. for all the material to be supplied.
 - Simulation of operation.
 - Verify interfaces connection ports to be installed.
 - Verification of redundancy of devices and network.
 - Verification of device failure scenarios for system robustness.
 - Verification of power failure and test all components when system is operating on each incoming power. Confirm no loss of functionality occurs.
 - Verification of fail-safe scenarios.
 - Verification that all alarms and interlocks work as specified.
 - 100% test of point, component, and function.
 - Review of the complete specified documentation package.
 - Review of equipment conformance with relevant codes and guidelines.

11.2.2 In addition, at the control panel factory:

- a. Panel fabricator shall conduct Pre-Test (or internal test) prior to the FAT with the following:
 - Perform continuity test of wiring to verify conformance of wiring circuits to approved panel wiring drawings. Correct discrepancies prior to FAT.

- Power up electronic components and operate continuously for 96 hours.
 - Provide a written verification of the tests showing:
 1. Dates and times of tests.
 2. Personnel performing tests.
 3. Discrepancies found.
 4. Date and person performing corrections.
 - b. The FAT shall be with the following:
 - Check individual software programs in each panel.
 - Check for I/O wiring completeness.
 - Check for PLC-to-PLC communications.
 - Check for remote terminal communications.
 - Calibrate furnished instrumentation.
 - c. During FAT, panel fabricator shall provide persons to conduct the test, shall provide staging area, for the time and space required, dedicated to the project in the panel shop for the testing, shall provide temporary network cabling to each panel, shall provide temporary power connection for both sources of panel power.
- 11.2.3 In addition, for PLC and SCADA program, simulation testing shall be provided to demonstrate operation of Control software and graphic screens with checklist as per SOO.
- The off-line test shall be in the setup as what we have on-site such as at least, not limited to, redundant PLC, Remote I/O, and VFD. The simulation testing shall also include scenario of comm lost (with VFD or Remote I/O or Secondary PLC), download event, local/remote operations.
- 11.2.4 All System Integrator's internal testing before the FAT should be documented properly according to the System Integrator internal procedures. Tests reports should be available before Micron's arrival at Vendor shop for FAT.
- 11.2.5 The tests shall be carried out at Vendor's expense.
- 11.2.6 If the system does not meet and perform in accordance with Vendor warranty obligation, Vendor will, at its expense, implement all the necessary modifications and afterwards the tests shall be repeated.
- 11.2.7 After Micron approval, the system will be prepared and shipped to the site.
- 11.2.8 The complete set of internal testing documents must be available before FAT. One copy shall be sent to Micron before FAT. No FAT shall be carried out without documents availability.

11.3 Installation Inspection

- 11.3.1 Once the control panel and relevant field devices have been totally installed and connected, installation inspection activities shall be carried out for Micron approval before the FMCS commissioning can be performed.
- 11.3.2 The panel passes a visual inspection and complies with the following:
 - a. Outer enclosure is labeled appropriately with ID tag.
 - b. Devices and modules are installed per the drawings.
 - c. Wiring is gathered with tie-wraps or bundled within wire-mold channels.
 - d. Wiring, devices and modules are labeled within the panel in accordance with the drawings.
 - e. Wiring terminations are adequate and firm, and wiring utilizes the color code established within the drawings.
 - f. Wiring cable shields are terminated at one end to power supply ground as appropriate.
 - g. Confirm panel and equipment rack are grounded to appropriate grounding source.
- 11.3.3 Devices are mechanically fastened properly, securely and firmly on mounting supports.
- 11.3.4 Field cable and sensor installation shall be visually inspected for physical damage and proper connection as per project specification and I/O list.
- 11.3.5 In general, verify identification and labelling at required locations for visibility, condition, legibility, and accuracy.
- 11.3.6 On all control and monitoring wiring, continuity check shall be carried out from primary element to PLC/RIO I/O interface. Under no circumstances will any control loop be energized without first performing continuity checks

11.4 SAT (Site Acceptance Test)

- 11.4.1 The final acceptance at site will be carried out after mechanical completion of the system at Micron facility. The purpose of the SAT is to verify that the system is supplied and installed and that it operates according to the relevant specification.
- 11.4.2 At the completion of installation of panels and field devices, each system to be tested by the System Integrator to show the following:
- a. Confirm input voltage (120Vac or 240Vac) is within appropriate tolerance range, and that the panel indicates normal operating condition upon the 120 or 240 Vac power application
 - b. Network Verification Testing, at the completion of network installation, control panels and their communication wiring are to be tested by the System Integrator to show the following:
 - The system is properly calibrated and adjusted in accordance with manufacturer's recommendations.
 - The network lines are free from grounded or open circuits. The control equipment will indicate when a ground or open circuit occurs.
 - Simulate power failure and test all components when system is operating on essential power. Confirm no loss of functionality occurs.
 - Where fiber optic cable is used, use an Optical Time Domain Reflectometer (OTDR) to document attenuation (optical power loss) in both directions from the ends of each fiber cable.
 - Where ethernet networks are used for FMCS, document the following using network testing means:
 - Network bandwidth throughput – maximum rate at which frames can be transmitted from the source to the destination with zero lost frames or errors.
 - Network Latency – the total time it takes for a frame to travel from source to destination. The minimum, average, and maximum latency values for each frame size shall be reported.
 - Confirm all exterior-mounted FMCS ethernet network components are equipped with appropriate surge protection.
 - The communication and device operation within each panel, between each panel and the head-end controls, and among all panels and the-end controls as an entire system is established and verified.

- The appropriate communication and alarm annunciation are demonstrated upon a test simulation program being loaded and executed.
- c. Correct operation and indication of grounds, shorts, and opens is tested at every point with the final operating software program.
- d. After continuity checks have been performed, the instrument loop will be energized. The point-to-point test must take place online from the sensor to the SCADA system.
 - Control valves will be stroked from the SCADA to verify correct action and operation at 0%, 25%, 50%, 75% and 100%.
 - Analog measurement shall be verified from the SCADA at 0%, 25%, 50%, 75%, and 100%.
 - All adjustable field devices such as trip switches, alarm switches, etc will be checked for correct operation of the loop.
 - The test report shall have the following, but not limited to :
 - System/Package Number/Name
 - Device Tag
 - PLC/RIO Panel Drawing Number/Name
 - PLC/RIO Panel Tag
 - All Alarm test
 - Test equipment reference and calibration certificate number.
 - If calibration is required, the calibration must take place online from the sensor to the SCADA. A calibration is a comparison of measuring equipment against a standard instrument of higher accuracy to detect, correlate, adjust, rectify and document the accuracy of the instrument being compared.
 - Sample test forms are shown in Appendix C for reference.
 - Perform SAT functional test in the final acceptance test procedure that is based on specified control operation in the system Sequence of Operation. Functional test shall be carried out from sensor, PLC, and actuator/final element, with SCADA to monitor all the function. The final acceptance test to be witnessed by Micron, or an authorized representative of Micron, Main Contractor, and System Integrator.

- Defects to be corrected at once and the test reconducted at no additional cost to Micron until the defect is corrected.

11.5 Documentation

- 11.5.1 Quality Management System should be in place and used for documentation of the commissioning activities.
- 11.5.2 The Vendor should provide a document list of his standard documentation and commissioning documentation package together with examples for review and final approval by Micron including a description of the vendor internal quality system.
- 11.5.3 Prior to the execution of any commissioning activity of the vendor, Micron is to be supplied with the respective protocols and test specification for approval – these must be made available to Micron at least 4 weeks before execution of the testing.

12.0 Document Control

12.1 **Approval:** GLOBAL_FAC_GFTT_APPR

12.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

12.3 Final Viewer Access

All Micron sites Facilities Team Members

12.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

12.5 Review

Biennially (two years)

13.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	12/08/20
1	2.2.10	Added: FMCS SCADA system integrator shall be Wonderware certified system platform system integrator with minimum past 5 years Wonderware system platform project experience	07/2/21
	11.2.2	Deleted: Calibrate panel fabricator furnished instrumentation.	
	11.2.3	Added: The off-line test shall be in the setup as what we have on-site such as at least, not limited to, redundant PLC, Remote I/O, and VFD. The simulation testing shall also include scenario of comm lost (with VFD or Remote I/O or Secondary PLC), download event, local/remote operations.	
	Appendix C Valve Control Loop Test Form	Replaced "Motor" with "Valve". Deleted "and SCADA feedback value".	
3	Section 3	Removed definitions for FCT, GFTT, Global Facilities Central , Construction, Operations, Planning teams	5/17/24
	Section 4	Removed GFTT and FCT Lead (Discipline Engineer) roles and replaced with Global Facilities Lead (Discipline Engineer) Changed GFTT Lead to both Global Engineering and Regional Lead for Site Project Construction Teams (responsibilities)	
	Appendix C	Added: Cable Termination Check in Instrument Loop Test Form	

Global Facilities – General - Management of Change (MOC) Procedure

Last Modified: 5/17/24

Version: 3

ID: A3YRXSD74VDV-57553043-401

Appendix A : Deviation List

Any deviation from this specification shall be submitted for approval during vendor evaluation. Deviation list format shall be the following:

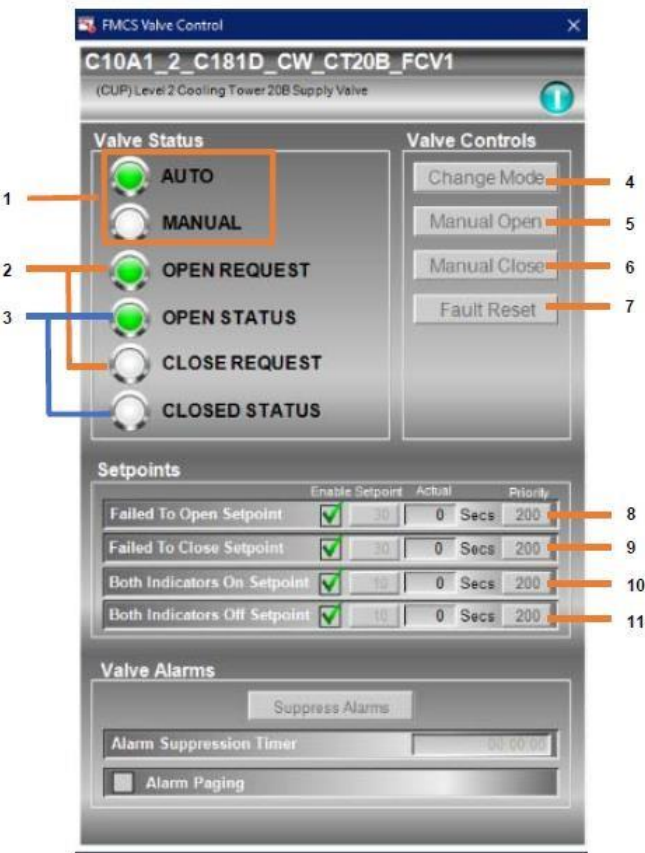
Site: _____

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT

Appendix B: SCADA Graphic Faceplate

The following faceplates shall be best practice to use:

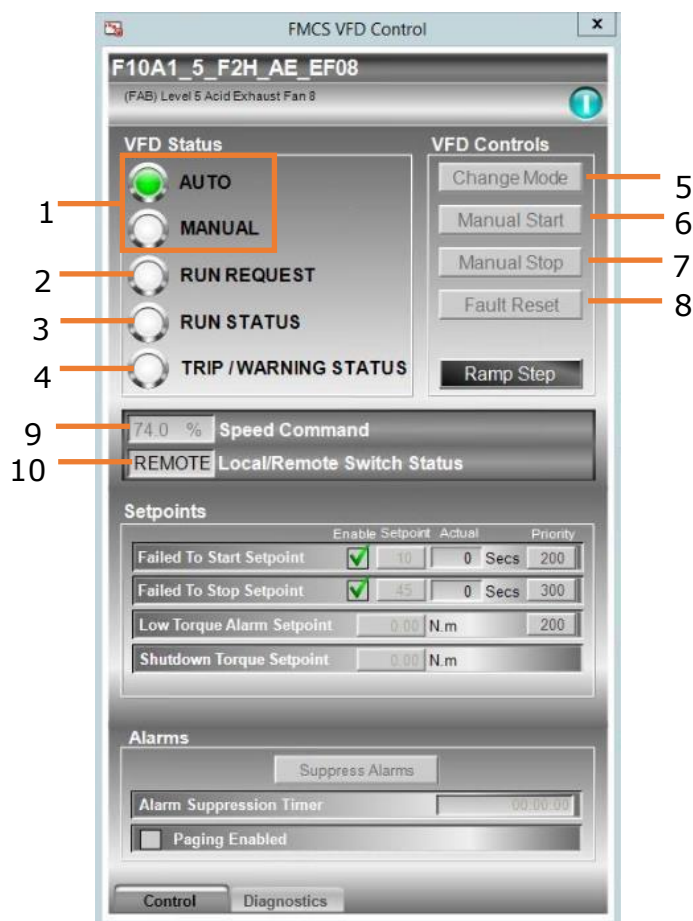
- 1. On/Off Valve or Damper Control



No	Section	Details
1	Auto/Manual (Status)	When valve in Remote mode, it can be Remote Manual or Remote Auto.
2	Open/Close Request (Status)	The status light is on at open/close request when the PLC send open/close command to the valve (remote manual or remote auto).
3	Open/Closed Status	The status light is on when received Open or Close status from valve feedback (either one).
4	Change Mode (Control)	Click to change Remote Manual to Remote Auto and vice versa (only in remote mode).
5	Manual Open (Control)	Click to open the valve (in Remote Manual)

6	Manual Close (Control)	Click to close the valve (in Remote Manual).
7	Fault Reset	Click to clear the fault alarm (Trip/Fail to Start / Fail to Stop)
8	Failed to Open Setpoint	Failed to open setpoint enable, setpoint limit and priority.
9	Failed to close setpoint	Failed to close setpoint enable, setpoint limit and priority.
10	Both indicators On setpoint	Both indicators On setpoint enable, setpoint limit and priority.
11	Both indicators Off Setpoint	Both indicators Off setpoint enable, setpoint limit and priority.

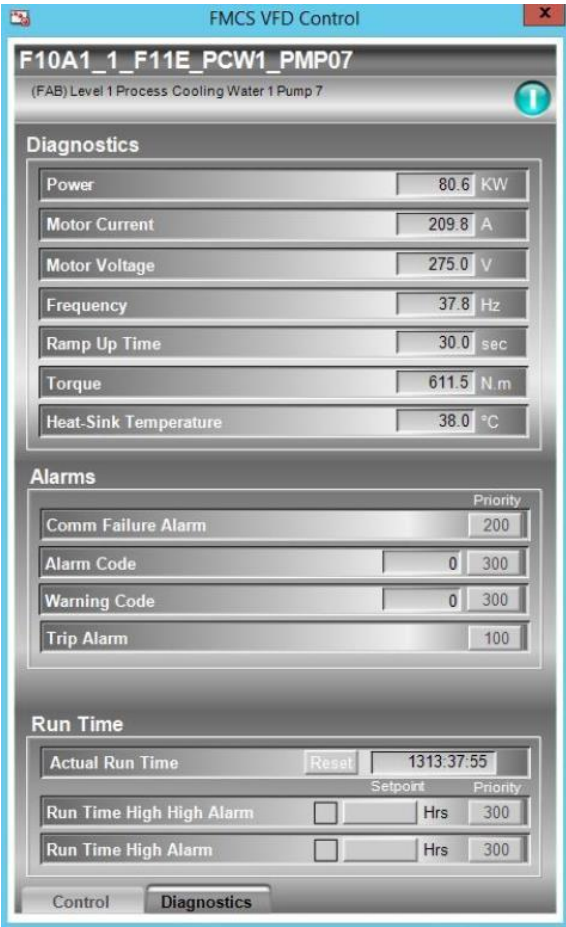
2. VFD Control



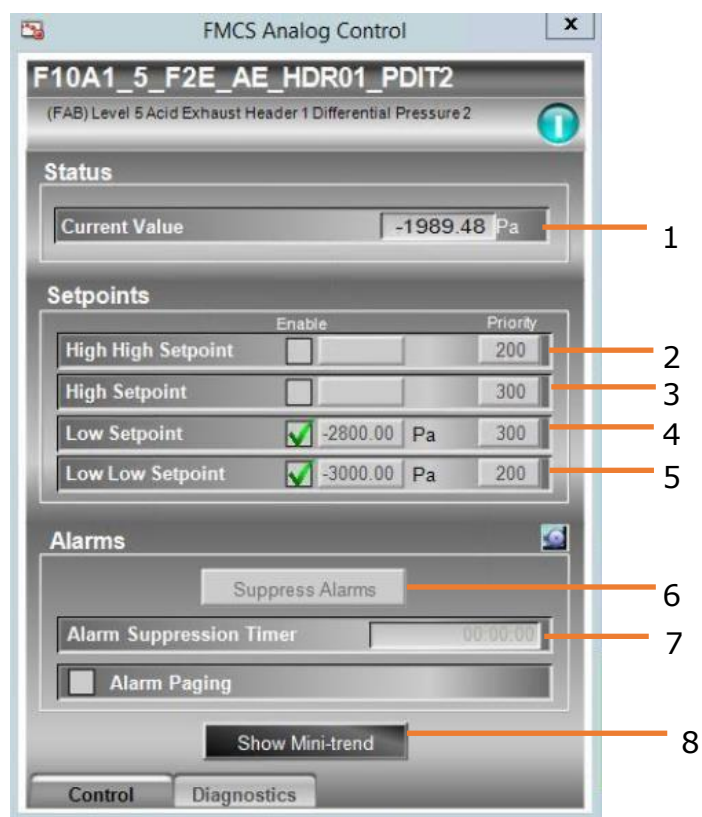
No	Section	Details
1	Auto/Manual (Status)	When VFD in Remote mode, it can be Remote Manual or Remote Auto.
2	Run Request (Status)	The status light is on when the PLC send run command to the VFD (remote manual or remote auto).
3	Run Status	The status light is green (Singapore, Malaysia, US, Japan) or red (China, Taiwan) when

		received run status from VFD
4	Trip/Warning Status	The status light is on if the VFD Trip or Fail to Start or Fail to Stop
5	Change Mode (Control)	Click to change Remote Manual to Remote Auto and vice versa (only in Remote mode)
6	Manual Start (Control)	Click to start the VFD (in Remote Manual)
7	Manual Stop (Control)	Click to stop the VFD (in Remote Manual)
8	Fault Reset	Click to clear the fault (Trip / Fail to Start / Fail to Stop)
9	Speed Command (%)	Key in the speed command in range of 0% to 100% (Remote Manual)
10	Local/Remote Switch Status	Remote or Local mode (Status)

VFD High Level Monitoring

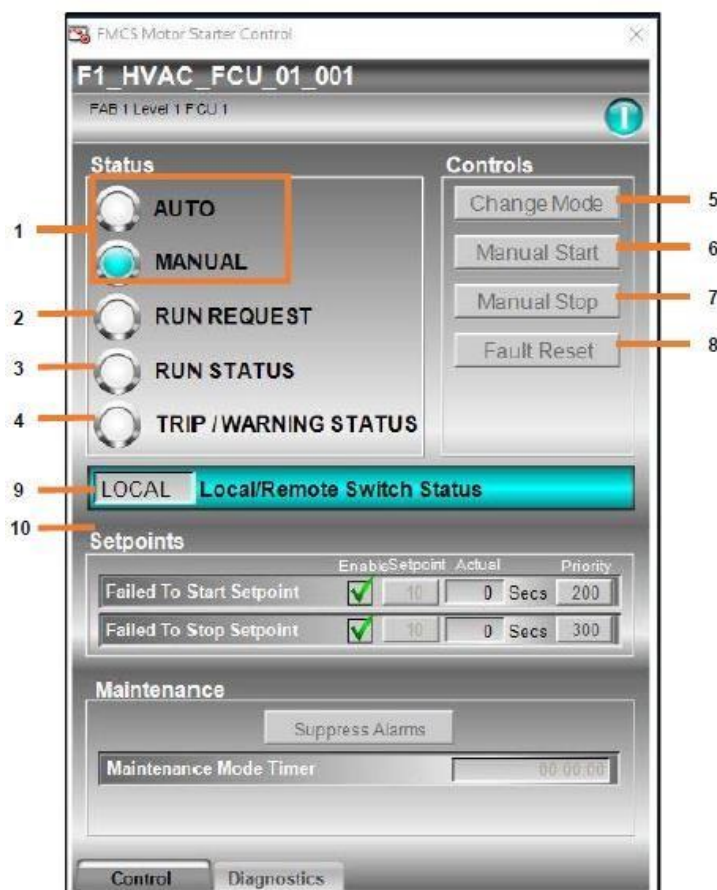


3. Analog Control



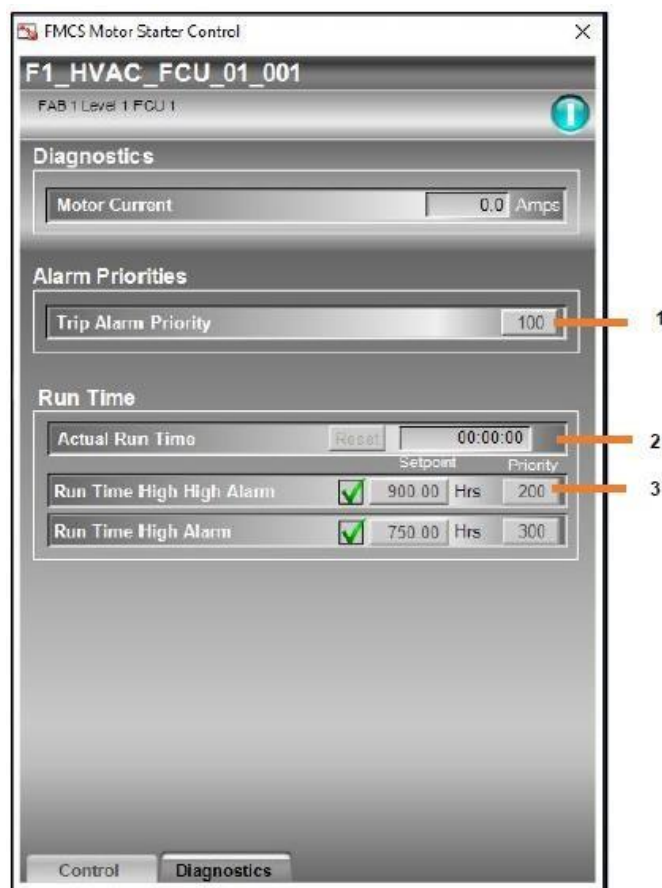
No	Section	Details
1	Current Value	Show the current value and unit for the analog value
2	High-High Setpoint	High-High alarm enable and setpoint limit
3	High Setpoint	High alarm enable and setpoint limit
4	Low Setpoint	Low alarm enable and setpoint limit
5	Low-Low Setpoint	Low-Low alarm enable and setpoint limit
6	Suppress Alarms	Temporary disable alarm for maintenance or troubleshoot
7	Alarm Suppression Timer	Timer for temporary disable alarm
8	Show Mini-trend	To show mini trend for this equipment

4. DOL Motor Control



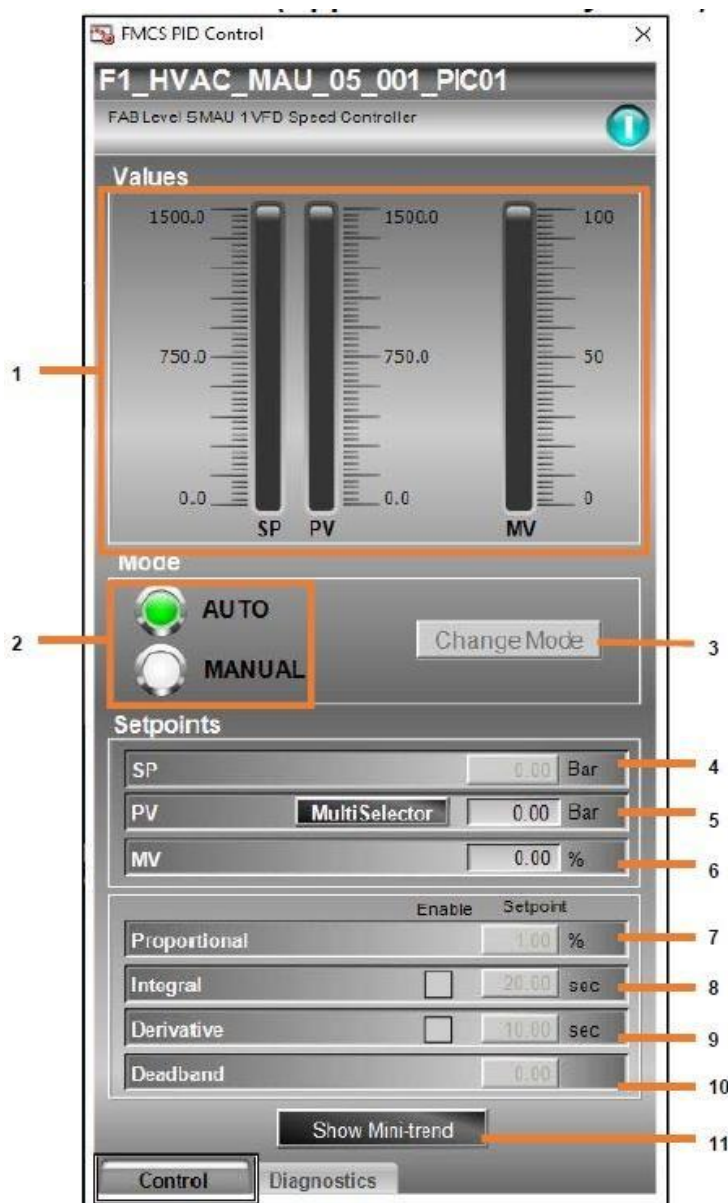
No	Section	Details
1	Auto/Manual (Status)	When Motor is Remote mode, it can be Remote Manual or Remote Auto
2	Run Request (Status)	The status light is on when the PLC send run command to the Motor (in remote manual or in remote auto)
3	Run Status	The light is green (for Singapore, Malaysia US, Japan) or red (for China, Taiwan) when received run status from Motor.
4	Trip/Warning Status	The status light is activated if the motor Trip or Fail to Start or Fail to Stop
5	Change Mode (Control)	Click to change Remote Manual to Remote Auto and vice versa (only in remote mode)
6	Manual Start (Control)	Click to start the motor (in Remote Manual)

7	Manual Stop (Control)	Click to stop the motor (in Remote Manual)
8	Fault Reset	Click to clear the fault (Trip / Fail to Start / Fail to Stop)
9	Local/Remote Switch Status	Remote or Local mode (status)
10	Fail to Start setpoint / Fail to Stop setpoint and Priority	Fail to start / stop alarm time delay and alarm priority



No	Section	Details
1	Alarm Priorities	Alarm priority setting
2	Run Time	Equipment Run Time
3	Run Time High-High Alarm Run Time High Alarm	Equipment Run Time alarm setting

5. PID Control



No	Section	Details
1	Graphic Value for SP, PV, and MV	Show the current value and unit for Setpoint (SP), Process Value (PV) and Manipulated Value (MV) in bar chart
2	Auto/Manual (Status)	PID mode, PID calculation shall run to modulate its output if it is in Auto, PID output shall be fixed if it is in Manual.
3	Change Mode (Control)	Click to change PIC mode from Manual to Auto and vice versa.

4	SP setpoint	To set value for Setpoint
5	PV setpoint	To set value for Process Value
6	MV setpoint	To set value for Manipulated Value
7	Proportional	To set value for Proportional parameter
8	Integral	To enable and set value for Integral parameter
9	Derivative	To enable and set value for Derivative parameter
10	Deadband	To set value for PID Deadband
11	Show Mini-Trend	To show mini trend for this PID values

Appendix C: Full Loop Test Forms

The following forms are recommended practice to follow:

- Instrument Loop Test Form
- Valve Control Loop Test Form
- Motor Control Test Form

INSTRUMENT LOOP TEST FORM

System/Package		PLC/RIO Panel Drawing	
Device Tag		PLC/RIO Panel Tag	

	Case	Description	Pass	Fail	N/A
1.	Are all instruments / Field Devices tagged correctly as per panel drawing ?				
2.	Instrument Range	Minimum: Max: Unit:.....			
3.	Measurement at the field instrument tally with the value shown at SCADA	Record data in table below			

Analog Input				Analog Output			
%Range	Field Device	HMI Value	SCADA Value	% Range	HMI Input	SCADA Value	Field Device Position
0				0			
25				25			
50				50			
75				75			
100				100			

Digital Input			Digital Output		
Field Device Input	HMI Value	SCADA Value	HMI Input	SCADA Value	Field Device Status

Alarm Test

	Test	Pass	Fail	N/A
1.	Bad I/O condition from field, Bad I/O received on SCADA			
2.	Default Value during Bad I/O : _____			

3.	Simulate Alarm for Analog Instruments based on the changing Alarm Limits on SCADA, the Alarms shall show at the SCADA.			
----	--	--	--	--

Cable Termination Check				
S/N	Responsible Site Personal to perform the following checks in the field and document the results as Pass, Fail or N/A.	Pass	Fail	N/A
1.	Comply cable size, insulation type as per data sheet/specification or cable schedule (inside & outside panel and indoor and Outdoor requirement)			
2.	Cable Gland - Confirm Size, Type is as per specification /area classification (If Any) and it is with complete Accessories.			
3.	Cable Gland - Confirm the tightness of glands and cable/wire and arranged neatly inside and outside of Panel, Instruments, JB, Actuators.			
4.	Ensure that insulated ferrule is not overlap by cable labelling			
5.	Confirm that the cable terminal point is properly secured into the terminal block with ferrule fully inserted.			
6.	Conduct cable pull out test one by one and check if it is firmly connected.			
7.	Confirm Termination box and Gland plates are properly closed without a gap.			
8.	Confirm Earth cable connected to body of equipment, panels, etc.			
9.	Confirm the Cable tag, terminal numbering and core cable marking are correct (at both ends) as per the Control Panel Wiring Diagram, RIO Panel wiring drawing, JB wiring drawing, Loop Drawing (if any).			
10.	Confirm cable insulation is undamaged, unexposed and proper insulation at crimping point.			
11.	Confirm the correct size and type of Cable lug has been used and crimped correctly.			
12.	Confirm the cable Continuity test - Passed.			
Note: If any Modification / Re-termination works done after final inspection, need to submit new Check Sheet for inspection. For installation and testing of special cable such as co-axial fibre optic, etc: refereeing shall be made to any special procedure or instruction that may be specified or agreed with relevant system suppliers.				

Comments: (If there are NO ADDITIONAL remarks enter NONE, initial and date)
Reason to be written in comments section and, where applicable, raised as an exception or outstanding issue as part of the system handover. If resolved prior to handover comments to be marked appropriately and signed of accordingly.

Test Equipment

Test equipment reference		Serial No:		Range:	
Certification Trace no:		Issue date:		Due Date:	

Comments: (If there are no comments, enter NONE, initial and date)

	Tested by	Witnessed by	Approved by
Company			
Signature			
Name			
Date			

VALVE CONTROL LOOP TEST FORM

System/Package		PLC/RIO Panel Drawing	
Equipment Tag		PLC/RIO Panel Tag	

	Case	Description	Pass	Fail	N/A
Control and Monitoring Status Test					
1.	SCADA command open	Valve shall open			
2.	SCADA command close	Valve Motor shall close			
3.	Is Direction of Action correct?				

Alarm Test

4.	Fail to Open	SCADA command Open, but receive no Open feedback			
5.	Fail to Close	SCADA command Close, but receive no Close feedback			
6.	Trip	SCADA receive Trip signal from valve			
7.	Valve Fail Position if Control Signal Lost	☐ Open ☐ Last Position ☐ Close			
8.	Valve Fail Position if Power Lost	☐ Open ☐ Last Position ☐ Close			
9.	Valve Fail Position if Pneumatic air lost	☐ Open ☐ Last Position ☐ Close			
10.	When bad I/O occurs from field, bad I/O received on SCADA				

Apply to only Modulating Control Valve

11.	Valve feedback tally with the Valve control command by SCADA in manual mode	Record data at table below			

Valve Feedback				Valve Com mand			
%Range	Valve Position	HMI Value	SCADA Value	% Range	HMI Input	SC .DA Value	Valve Command Signal
0				0			
25				25			
50				50			
75				75			
100				100			

Comments: (If there are no comments, enter NONE, initial and date)

	Tested by	Witnessed by	Approved by
Company			
Signature			
Name			
Date			

MOTOR CONTROL LOOP TEST FORM

System/Package		PLC/RIO Panel Drawing	
Equipment Tag		PLC/RIO Panel Tag	

			Pass	Fail	N/A
	Case	Description			
Control and Monitoring Status Test					
1.	Local mode	Motor is at local mode and SCADA shows it is in local mode			
2.	Remote mode	Motor is at remote mode and SCADA shows it is in remote mode. Local start/stop pushbutton shall not be able to function.			
3.	SCADA command start in Remote mode	Motor shall run			
4.	SCADA command stop in Remote mode	Motor shall stop			
5.	Motor Run-Hours	SCADA shows and update Run hours			
Alarm Test					
6.	Fail to Run	SCADA command start in Remote mode, but receive no run feedback			
7.	Fail to Stop	SCADA command stop in Remote mode, but still receive run feedback			
8.	Trip	SCADA receive Trip signal from motor/equipment			
9.	When bad I/O occurs from field, bad I/O received on SCADA				
Apply to only Motor with VFD					
10.	VFD Frequency scaling is set correctly	Minimum:Hz Maximum:Hz			

11.	VFD frequency feedback at the equipment tally with the frequency shown at SCADA	Record data at table below			
12.	In Remote Mode, VFD frequency shall tally with the SCADA frequency command	Record data at table below			

Apply to only Motor with VFD							
Frequency Feedback				Frequency Command			
%Range	Field VFD	HMI Value	SCADA Value	% Range	HMI Input	SCADA Value	Field VFD
0				0			
25				25			
50				50			
75				75			
100				100			

Other Monitoring Data					
Data Point	At VFD	At SCADA	Pass	Fail	N/A

Comments: (If there are no comments, enter NONE, initial and date)

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	Tested by	Witnessed by	Approved by
Company			
Signature			
Name			
Date			

Appendix D: Green Tag and Blue Tag Commissioning Forms

The following forms are recommended practice to follow:

- Green Tag Commissioning Form
- Blue Tag Commissioning Form

Last Modified: 5/17/24

Version: 3

ID: A3YRXSD74VDV-57553043-401

GREEN TAG FORM

[illegible]

Green Tag Form.xlsx

BLUE TAG FORM

Project:		Date:		Completed by System Owners/Operations							Completed by Operations/System Integration						
System Owners:				Alarm Description (if different from Tag Description)	Discrete Input: Alarmed Y/N?	Analog Alarm Setpoint Values (N/A if not used)					Time Delay Duration (sec)	Alarm Suppression Off	Alarm Verified	Trending Submitted	Trending Active	Comments	Initials
Tagname	Range	Tag Description	LOLO			LO	HI	HIHI	DB								



Blue Tag Form.xlsx